

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

کیا آپ نے آج درود شریف پڑھا

نہیں تو ابھی پڑھ لیجئے

صلی اللہ علیہ وسلم



VU Stars

اسلام علیکم:

امید کرتا ہوں سب خیریت سے ہوں گئے۔ آنے والے تمام نیو طالب علم کو میں ورچوئل یونیورسٹی آف پاکستان میں خوش آمدید کرتا ہوں۔
اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابطہ کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ،
فیس پے اور چینج سٹڈی پروگرام وغیرہ۔

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اگر کوئی سٹوڈنٹ کسی مجبوری کی وجہ سے یا جاب ہو لڈر ہونے کی وجہ سے اپنا ایل ایم ایس خود ہینڈل نہیں کر سکتا تو وہ ہم سے پیڈ سروس حاصل کر
سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹھیمپٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

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مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔

لوح قرآنی

ن

حَمْدُكَ

الْحَمْدُ

لَیْسَ

عَمَّ

الْمَصَّ

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كَلَّعَصَّ

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اس کے دیکھنے والوں کی سب مشکلیں آسان ہو جائیں گی صبح اس کو دیکھ کر جو کام شروع کیا جائیگا پورا ہوگا اور غیبی طریقوں سے رزق آنے لگے گا!

Question # 2 of 10 (Start time: 07:49:49 PM, 17 January 2024)

Total Marks: 1

How many select lines would be required for an 8x1 multiplexer?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 8 |
| ☐ | 1 |
| ☐ | 3 |
| ☐ | 2 |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 07:50:04 PM, 17 January 2024)

Total Marks: 1

Which of the following symbols is used to represent XOR operation?

Select the correct option

[Reload Math Equations](#)

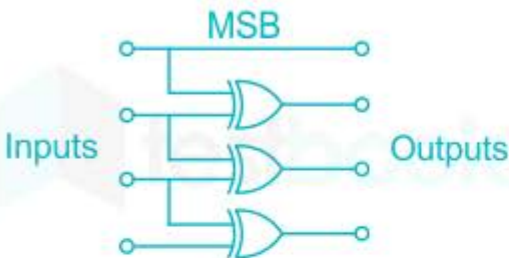
☐	$\oplus$
☐	$\times$
☐	$\wedge$
☐	$+$

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 5 of 10 (Start time: 07:51:29 PM, 17 January 2024)

Total Marks: 1

Following diagram is of _____ code conversion.



Select the correct option

[Reload Math Equations](#)

☐	BCD
☐	ASCII
☐	Excess-3
☐	Gray

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1



Choose Answer

④

Question # 6 of 10 (Start time: 07:52:15 PM, 17 January 2024)

Total Marks: 1

Convert the Gray code 1011 to binary.

Select the correct option

[Reload Math Equations](#)

☐	1011
☐	0111
☐	1101
☐	1010

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 (Start time: 07:52:32 PM, 17 January 2024)

Total Marks: 1

A K-Map of size 2x2 can be used to map a _____ input boolean expression.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 1 |
| ☐ | 4 |
| ☐ | 2 |
| ☐ | 3 |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Total Marks: 1


$$A \oplus B \oplus C_{in}$$
$$(A \times C_{in}) \oplus B$$
$$(A \times B) \oplus C_{in}$$
$$(A + B) \times C_{in}$$

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1



- Show Answer

Total Marks: 1

To create a Johnson Counter by using D-flip flops, how flip flops should be connected together?



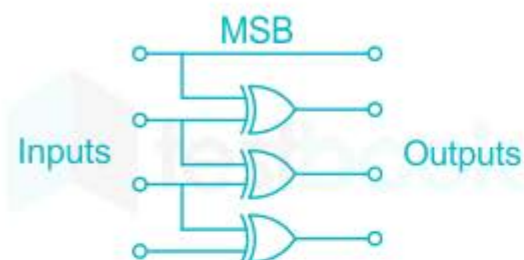
- Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 10 (Start time: 07:56:31 PM, 17 January 2024)

Total Marks: 1

Following diagram is of _____ code conversion.



Select the correct option

[Reload Math Equations](#)

☐	BCD
☐	ASCII
☐	Excess-3
☐	Gray

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 10 (Start time: 07:56:37 PM, 17 January 2024)

Total Marks: 1

When implementing a half adder by using gates, which of the following gates is used to return carry bit as an output?

Select the correct option

[Reload Math Equations](#)

☐	AND
☐	NAND
☐	XOR
☐	OR

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 07:56:47 PM, 17 January 2024)

Total Marks: 1

Reflected Binary Code is also known as _____ code.

Select the correct option

[Reload Math Equations](#)

☐	Excess-3
☐	Gray Code
☐	BCD
☐	ASCII

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 07:56:47 PM, 17 January 2024)

Total Marks: 1

Reflected Binary Code is also known as _____ code.

Select the correct option

[Reload Math Equations](#)

☐	Excess-3
☒	Gray Code
☐	BCD
☐	ASCII

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 4 of 10 (Start time: 07:57:13 PM, 17 January 2024)

Total Marks: 1

In a full adder circuit having inputs **A = 1, B = 1, Cin = 0**, the outputs **S** and **Cout** will be _____.

Select the correct option

[Reload Math Equations](#)☐

S = 0, Cout = 0

☐

S = 1, Cout = 1

☐

S = 0, Cout = 1

☐

S = 1, Cout = 0

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1



- Show Answer

Total Marks: 1

Reload Math Equations

Show Answer

Total Marks: 1



 Reload Math Equations

Registers

K-maps

Latches

Terms

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 (Start time: 07:58:19 PM, 17 January 2024)

Total Marks: 1

When implementing a half adder by using gates, which of the following gates is used to give Sum (S) as an output?

[Reload Math Equations](#)

☐	AND
☐	OR
☐	XOR
☐	NOR

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 8 of 10 (Start time: 07:58:29 PM, 17 January 2024)

Total Marks: 1

Each group of adjacent Minterms (group size in powers of twos) corresponds to a possible product term of the given _____

Select the correct option

 Reload Math Equations

☐	Set
☐	Word
☐	Value
☐	Function

Show Answer

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 07:58:37 PM, 17 January 2024)

Total Marks: 1

A decimal counter has _____ states.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|----|
| ☐ | 10 |
| ☐ | 2 |
| ☐ | 7 |
| ☐ | 9 |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 10 of 10 (Start time: 07:58:45 PM, 17 January 2024)

Total Marks: 1

Each product term of a group, $w'xy'$ and wy , represents the _____ in that group.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|-----------------|
| ☐ | Input |
| ☐ | Sum of Maxterms |
| ☐ | POS |
| ☐ | Sum of Minterms |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 1 of 10 (Start time: 08:17:36 PM, 17 January 2024)

Total Marks: 1

A counter circuit is usually constructed of _____

Select the correct option

[Reload Math Equations](#)☐

A number of NOR gates connected in cascade form

☐

A number of latches connected in cascade form

☐

A number of NAND gates connected in cascade form

☐

A number of flip-flops connected in cascade

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 10 (Start time: 08:18:31 PM, 17 January 2024)

Total Marks: 1

Which of the following expressions represents **S** output in a full adder?

[Reload Math Equations](#)

Select the correct option

☐

$$(A \times B) \oplus C_{in}$$

☐

$$(A + B) \times C_{in}$$

☐

$$A \oplus B \oplus C_{in}$$

☐

$$(A \times C_{in}) \oplus B$$

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 08:18:51 PM, 17 January 2024)

Total Marks: 1

How many select lines would be required for an 8x1 multiplexer?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 3 |
| ☐ | 1 |
| ☐ | 8 |
| ☐ | 2 |

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1



- Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 4 of 10 (Start time: 08:19:01 PM, 17 January 2024)

Total Marks: 1

Three decade counter would have _____

Select the correct option

[Reload Math Equations](#)☐

2 BCD counters

☐

4 BCD counters

☐

3 BCD counters

☐

5 BCD counters

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 5 of 10 (Start time: 08:19:43 PM, 17 January 2024)

Total Marks: 1

Reflected Binary Code is also known as _____ code.

Select the correct option

[Reload Math Equations](#)☐

Excess-3

☐

Gray Code

☐

BCD

☐

ASCII

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 9 of 10 (Start time: 08:20:24 PM, 17 January 2024)

Total Marks: 1

What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

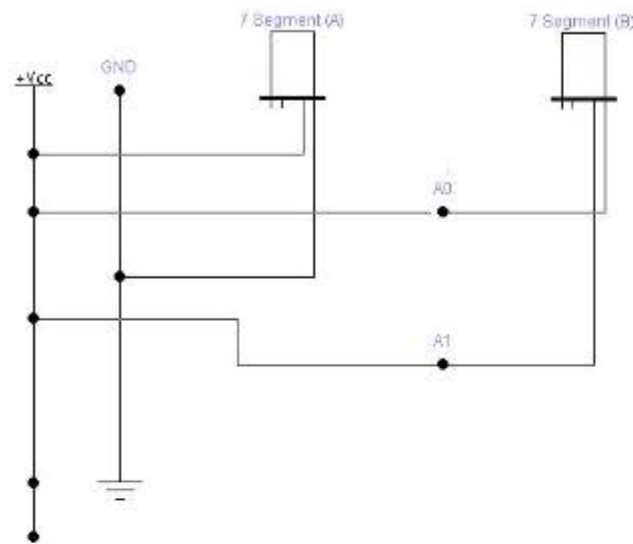
Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|----------------|
| ☐ | 0 to $2^n - 1$ |
| ☐ | 0 to 2^n |
| ☐ | 0 to $2^n + 1$ |
| ☐ | 0 to 2^{n+1} |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Consider the below given EWB circuit, what values will be displayed on 7-segment displays A and B when the circuit simulation is activated in EWB?



Select the correct option

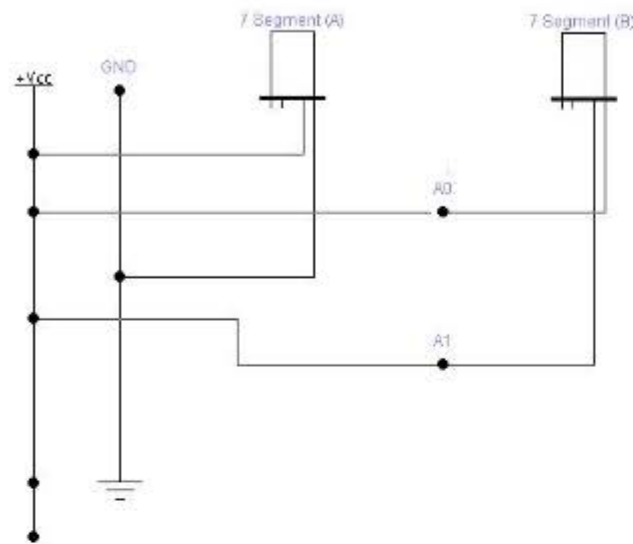
[Reload Math Equations](#)

Show Answer

- | | |
|-----------------------|------|
| ☐ | 1, 3 |
| ☐ | 2, 3 |
| ☐ | 0, 1 |
| ☐ | 3, 3 |

[Click to Save Answer & Move to Next Question](#)

Consider the below given EWB circuit, what values will be displayed on 7-segment displays A and B when the circuit simulation is activated in EWB?



Select the correct option

[Reload Math Equations](#)

Show Answer

- | | |
|-----------------------|------|
| ☐ | 1, 3 |
| ☐ | 2, 3 |
| ☐ | 0, 1 |
| ☐ | 3, 3 |

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 10 (Start time: 08:24:59 PM, 17 January 2024)

Total Marks: 1

A K-Map of size 2x2 can be used to map a _____ input boolean expression.

Select the correct option

[Reload Math Equations](#)

3



1



2



4

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 08:25:06 PM, 17 January 2024)

Total Marks: 1

Three decade counter would have _____

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|----------------|
| ☐ | 5 BCD counters |
| ☐ | 3 BCD counters |
| ☐ | 2 BCD counters |
| ☐ | 4 BCD counters |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 4 of 10 (Start time: 08:25:18 PM, 17 January 2024)

Total Marks: 1

Why do we use Gray Code?

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|------------------------------------|
| ☐ | Invert the code |
| ☐ | Error Correction |
| ☐ | To count the number or bit changes |
| ☐ | Error Detection |

Show Answer

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 08:26:37 PM, 17 January 2024)

Total Marks: 1

Reflected Binary Code is also known as _____ code.

Select the correct option

[Reload Math Equations](#)

☐	Excess-3
☐	Gray Code
☐	ASCII
☐	BCD

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 6 of 10 (Start time: 08:26:49 PM, 17 January 2024)

Total Marks: 1

The parallel outputs of a counter circuit represent the _____

Select the correct option

[Reload Math Equations](#)

☐	Parallel data word
☐	Clock frequency
☐	Clock count
☐	Counter modulus

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Total Marks: 1

Select the correct option

 Reload Math Equations

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1

When implementing a half adder by using gates, which of the following gates is used to give Sum (S) as an output?

 Reload Math Equations

- Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 10 of 10 (Start time: 08:27:39 PM, 17 January 2024)

Total Marks: 1

Total number of inputs in a half adder is _____.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 2 |
| ☐ | 4 |
| ☐ | 1 |
| ☐ | 3 |

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 5 (Start time: 02:33:08 PM, 18 January 2024)

Total Marks: 1

In min heap, deletemin() will put the _____ element at the first position of the array.

Select the correct option

[Reload Math Equations](#)

☐	Second
☐	Last
☐	None of the given options.
☐	Third

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 5 (Start time: 11:04:55 AM, 19 January 2024)

Total Marks: 1

How many flop flops are required to create a counter circuit which can count from 0 to 7?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 1 |
| ☐ | 3 |
| ☐ | 4 |
| ☐ | 2 |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 2 of 5 (Start time: 11:05:03 AM, 19 January 2024)

Total Marks: 1

To create a Johnson Counter by using D-flip flops, how flip flops should be connected together?

Select the correct option

[Reload Math Equations](#)

Q output of previous flip flop should be connected to clock input of second flip flop



Q output of previous flip flop should be connected to D input of second flip flop



Q' output of previous flip flop should be connected to clock input of second flip flop



Q' output of previous flip flop should be connected to D input of second flip flop

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 5 (Start time: 11:05:40 AM, 19 January 2024)

Total Marks: 1

The parallel outputs of a counter circuit represent the _____

Select the correct option

[Reload Math Equations](#)

☐	Counter modulus
☐	Parallel data word
☐	Clock frequency
☐	Clock count

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 4 of 5 (Start time: 11:05:49 AM, 19 January 2024)

Total Marks: 1

Three decade counter would have _____

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|----------------|
| ☐ | 4 BCD counters |
| ☐ | 5 BCD counters |
| ☐ | 3 BCD counters |
| ☐ | 2 BCD counters |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 5 of 5 (Start time: 11:05:57 AM, 19 January 2024)

Total Marks: 1

How many select lines would be required for an 8x1 multiplexer?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 8 |
| ☐ | 2 |
| ☐ | 1 |
| ☐ | 3 |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 1 of 5 (Start time: 04:34:19 PM, 24 January 2024)

Total Marks: 1

Reflected Binary Code is also known as _____ code.

Select the correct option

[Reload Math Equations](#)

☐	BCD
☐	Gray Code
☐	Excess-3
☐	ASCII

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 5 (Start time: 04:34:27 PM, 24 January 2024)

Total Marks: 1

What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|----------------|
| ☐ | 0 to 2^{n+1} |
| ☐ | 0 to 2^n |
| ☐ | 0 to $2^n + 1$ |
| ☐ | 0 to $2^n - 1$ |

Show Answer

[Click to Save Answer & Move to Next Question](#)

Total Marks: 1

Show Answer

3

Question # 4 of 5 (Start time: 04:34:33 PM, 24 January 2024)

Total Marks: 1

Comparators are used in _____

Select the correct option

 Reload Math Equations

☐	Hard Drive
☐	Memory
☐	CPU
☐	Motherboard

Show Answer

Click to Save Answer & Move to Next Question

Question # 5 of 5 (Start time: 04:34:40 PM, 24 January 2024)

Total Marks: 1

Which of the following is an invalid BCD code?

Select the correct option

[Reload Math Equations](#)

☐	0011
☐	1101
☐	0101
☐	1001

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Total Marks: 1



- Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 5 (Start time: 04:51:18 PM, 24 January 2024)

Total Marks: 1

What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|----------------|
| ☐ | 0 to 2^{n+1} |
| ☐ | 0 to 2^n |
| ☐ | 0 to $2^n + 1$ |
| ☒ | 0 to $2^n - 1$ |

[Show Answer](#)

Saving...

Question # 2 of 5 (Start time: 04:51:35 PM, 24 January 2024)

Total Marks: 1

TTL 74LS85 is a _ _ _ _ _

Select the correct option

 Reload Math Equations

8-bit word comparator



8-bit magnitude comparator



1-bit digital comparator



4-bit magnitude comparator

Show Answer

Click to Save Answer & Move to Next Question



Question # 2 of 5 (Start time: 04:51:35 PM, 24 January 2024)

Total Marks: 1

TTL 74LS85 is a _ _ _ _ _

Select the correct option

 Reload Math Equations

8-bit word comparator



8-bit magnitude comparator



1-bit digital comparator



4-bit magnitude comparator

Show Answer

Click to Save Answer & Move to Next Question



Total Marks: 1



- Show Answer

Question # 4 of 5 (Start time: 04:51:53 PM, 24 January 2024)

Total Marks: 1

Which of the following is the modified code of Excess-3 code?

Select the correct option

[Reload Math Equations](#)

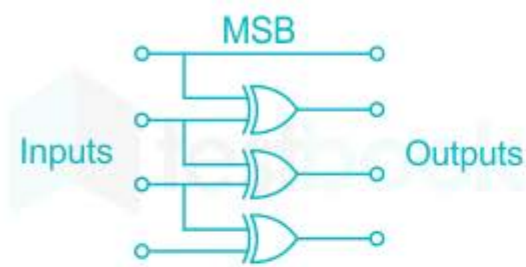
☐	BCD
☐	EBDIC
☐	Gray Code
☐	ASCII

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 5 of 5 (Start time: 04:52:02 PM, 24 January 2024)

Total Marks: 1

Following diagram is of _____ code conversion.



Select the correct option

[Reload Math Equations](#)

☐	ASCII
☐	BCD
☐	Gray
☐	Excess-3

Show Answer

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 5 (Start time: 06:38:38 PM, 24 January 2024)

Total Marks: 1

Three decade counter would have _____

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|----------------|
| ☐ | 2 BCD counters |
| ☐ | 3 BCD counters |
| ☐ | 4 BCD counters |
| ☐ | 5 BCD counters |

Show Answer

Click to Save Answer & Move to Next Question

Question # 2 of 5 (Start time: 06:38:46 PM, 24 January 2024)

Total Marks: 1

How many flop flops are required to create a counter circuit which can count from 0 to 7?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | 3 |
| ☐ | 1 |
| ☐ | 4 |
| ☐ | 2 |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

Question # 3 of 5 (Start time: 06:39:04 PM, 24 January 2024)

Total Marks: 1

What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|----------------|
| ☐ | 0 to $2^n - 1$ |
| ☐ | 0 to 2^n |
| ☐ | 0 to 2^{n+1} |
| ☐ | 0 to $2^n + 1$ |

[Show Answer](#)[Click to Save Answer & Move to Next Question](#)

14°C Smoke



ENG



6:39 PM

1/24/2024

3

Question # 4 of 5 (Start time: 06:39:16 PM, 24 January 2024)

Total Marks: 1

An identify comparator is defined as a digital comparator which has _____

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|--------------------------|
| ☐ | Two output terminals |
| ☐ | No output terminals |
| ☐ | Only one output terminal |
| ☐ | Three output terminals |

Show Answer

[Click to Save Answer & Move to Next Question](#)

14°C Smoke



ENG

6:39 PM
1/24/2024

Question # 5 of 5 (Start time: 06:39:26 PM, 24 January 2024)

Total Marks: 1

Convert the Gray code 1011 to binary.

Select the correct option

 Reload Math Equations

☐	1101
☐	1011
☐	0111
☐	1010

Show Answer

Click to Save Answer & Move to Next Question